LAB EXPERIMENT - 13

Department	AIDS
Course Code / Course Name	ITA1409 - ETHICAL HACKING
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Q13. Host Discovery

AIM

To perform host discovery using the Nmap tool in the Kali Linux operating system and to understand the different host discovery options used to find the live hosts in a network.

COMMANDS TO EXECUTE

Open the terminal and execute the following commands:

1. **Treat the Host as Online (-Pn)**

   The -Pn option skips the host discovery and treats the target as online, then scans its ports.

   Command: `nmap -Pn 192.168.1.1`

2. **Ping Scan (-sn)**

   The -sn option performs only host discovery using a ping scan and does not scan the ports of the target.

   Command: `nmap -sn 192.168.1.1`

3. **ARP Discovery (-PR)**

   The -PR option uses ARP requests to discover the hosts present in the local network.

   Command: `nmap -PR 192.168.1.1`

4. **Scan without DNS Resolution (-n)**

   The -n option disables the DNS resolution, which makes the scan faster.

   Command: `nmap -n 192.168.1.1`

OUTPUT:


┌──(root㉿kali)-[~]
└─# nmap -Pn192.168.1.1
Starting Nmap 7.93 (https://nmap.org) at 2023-01-30 01:15 EST
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.02 seconds
┌──(root㉿kali)-[~]
└─# nmap -sn192.168.1.1
Nmap 7.93 (https://nmap.org)
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
Can pass hostnames, IP addresses, networks, etc.
Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
-iL <inputfilename>: Input from list of hosts/networks
-iR <num hosts>: Choose random targets
--exclude <host1[,host2][,host3],...>: Exclude hosts/networks
--excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
-sL: List Scan - simply list targets to scan
-sn: Ping Scan - disable port scan
-Pn: Treat all hosts as online -- skip host discovery
-PS/PA/PU/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
-PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
-PO[protocol list]: IP Protocol Ping
-n/-R: Never do DNS resolution/Always resolve [default: sometimes]
--dns-servers <serv1[,serv2],...>: Specify custom DNS servers
--system-dns: Use OS's DNS resolver
--traceroute: Trace hop path to each host
SCAN TECHNIQUES:
-sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
-sU: UDP Scan
-sN/sF/sX: TCP Null, FIN, and Xmas scans
--scanflags <flags>: Customize TCP scan flags
-sI <zombie host[:probeport]>: Idle scan
-sY/sZ: SCTP INIT/COOKIE-ECHO scans
-sO: IP protocol scan
-b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
-p <port ranges>: Only scan specified ports
Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
--exclude-ports <port ranges>: Exclude the specified ports from scanning
-F: Fast mode - Scan fewer ports than the default scan
-r: Scan ports sequentially - don't randomize
--top-ports <number>: Scan <number> most common ports
--port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
-sV: Probe open ports to determine service/version info
--version-intensity <level>: Set from 0 (light) to 9 (try all probes)
--version-light: Limit to most likely probes (intensity 2)
--version-all: Try every single probe (intensity 9)
--version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:



RESULT

The host discovery options of the nmap command (-Pn, -sn, -PR and -n) were executed in Kali Linux. The scan of the target 192.168.1.1 using the -n option showed that the host is up with a latency of 0.0022 seconds, and all 1000 scanned ports were reported as filtered.